

VISHWA G S

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Executive Summary

Mechatronics Engineering student with hands-on experience in developing robotic systems using ROS and ROS2, including SLAM-based mapping, sensor fusion involving LiDAR, cameras, IMUs, and wheel encoders, and Nav2-based autonomous navigation, taking projects from simulation to physical hardware deployment. Proficient in Linux and Git-based workflows, with experience integrating multi-sensor perception data and debugging robot behavior in real-world environments. Seeking the Robotics Software Intern role at ANSCER Robotics to contribute to mobile robotics and fleet management software while further strengthening my skills in perception, planning, and hardware-software integration.

Technical Skills

- Robotics Frameworks: ROS2 (Humble), Nav2 Navigation Stack, SLAM Toolbox, TF2, URDF Modeling
- Simulation & Visualization: Gazebo, RViz
- Programming Languages: Python, C++
- Perception & Computer Vision: RPLiDAR integration, OpenCV, YOLOv8, Object Detection, Camera-to-Robot Coordinate Transformation
- Tools & Platforms: Git & GitHub, Ubuntu Linux
- Industrial Automation & IoT: PLC Programming, OPC UA, MQTT, Node-RED, ESP32

Soft Skills

- Problem Solving
- Team Collaboration
- Technical Documentation
- Communication
- Adaptability & Quick Learning
- Leadership
- Multilingual: English (Professional), Tamil (Native)

Professional Experience

Autonomous Mobile Robot (AMR) — Personal/Academic Project

February 2026 – Present (Ongoing)

Role: Project Developer

Responsibilities:

- Developed a differential-drive AMR using ROS2, covering the full development cycle from simulation to real-world hardware deployment.
- Designed a custom robot model in URDF and validated kinematics and behavior through Gazebo simulation.
- Integrated RPLiDAR for environmental perception and performed real-time mapping with SLAM Toolbox to generate occupancy grid maps.
- Upgraded the LiDAR sensor from LDS-01 to RPLiDAR A2 M8 to extend detection range and improve mapping accuracy.
- Configured the ROS2 ecosystem, including TF transformations, robot state publishing, and wheel encoder-based odometry for accurate pose estimation.
- Implemented autonomous navigation using the Nav2 stack, achieving reliable goal-based navigation and obstacle avoidance.

- Tested the robot on physical hardware, validating mapping/localization performance and debugging sensor and navigation issues in real-world conditions.
- Built a custom PyQt-based dashboard for robot monitoring, visualization, and control of live operations.

Impact:

- Delivered a complete ROS2-based AMR platform spanning simulation, perception, localization, and navigation.
- Built and deployed a custom UI that improved robot monitoring and operator interaction.
- Demonstrated end-to-end robotics deployment, sensor integration, and debugging on physical hardware.

MachineX3 — Industrial Coolant Monitoring System

July 2025 – September 2025 (Approx., 2 months)

Role: IoT & Automation Developer

Responsibilities:

- Developed an industrial IoT solution to monitor coolant and water levels in real time, integrating ultrasonic and flow sensors with ESP32 microcontrollers.
- Used a PLC alongside OPC UA communication instead of relay- or microcontroller-only control, aligning the system with industrial automation standards.
- Published sensor data via MQTT and designed a Node-RED dashboard for real-time monitoring, alarm/status alerts, and automated reporting via Google Sheets.

Impact:

- Enabled real-time visibility into industrial coolant usage and demonstrated practical integration of embedded systems with industrial automation infrastructure.

Object Detection and Coordinate Estimation for Robotic Manipulation

December 2025 – January 2026 (Approx., 20 days)

Role: Robotics & Computer Vision Developer

Responsibilities:

- Built a simulation-based robotic manipulation system that detects objects via camera and estimates their spatial (X, Y, Z) coordinates using computer vision techniques.
- Fed generated coordinates into robotic arm motion planning, validating camera-to-robot coordinate transformation across the full pick-and-place workflow.

Impact:

- Integrated perception output directly with robotic arm motion-planning logic, building foundational experience in vision-guided pick-and-place automation.

Education

Bachelor of Engineering (B.E.) — Mechatronics Engineering

Bannari Amman Institute of Technology | CGPA: 8.7 | Expected 2028

Certifications

- NPTEL Certification — Mechatronics

Awards & Accomplishments

- Displayed Autonomous Mobile Robot prototype at CODISSIA Intec.
- Won “Most Innovative Design” at Cadethon.
- Won First Prize at BITHACK Season 1.